3a	the <u>*increase* in internal energy</u> of a system is equal to the <u>sum</u> of the <u>*heat supplied to*</u> the system and the <u>*work done*</u> <u>*on*</u> the system. Only get 1 mark if candidate did not use the word “system”.
3b	At <u>constant volume</u>, <u>no work is done</u> by the gas. <u>Heat</u> is required to increase the temperature of unit mass of gas by one unit <u>is solely to increase its internal energy</u>. At constant pressure, work is done by the gas as it expands. <u>Heat is required for the increase of its internal energy and the work done by the gas</u>. Hence the specific heat capacity of ideal gas at constant pressure is greater than the specific heat capacity at constant volume.
3c	Internal energy consists of potential energy due to intermolecular forces and kinetic energy due to random motion. Temperature is only making reference to the kinetic energy due to random motion. Temperature is proportional to the average kinetic energy due to random motion and not the total kinetic energy of all the particles.
3di1	Equation of state: $pV = nRT$ $2.62\times10^{5}(0.0360) = n(8.31)(273.15+25)$ $n = 3.809\text{ mol} = 3.8\text{ mol}$

3di2	$pV = nRT$ $p(0.0360) = (0.30 + 3.809)(8.31)(273.15 + 25)$ $p = 2.827 \times 10^5$ Pa

4ai	amplitude is decreasing (very) gradually / oscillations would continue (for a long time)